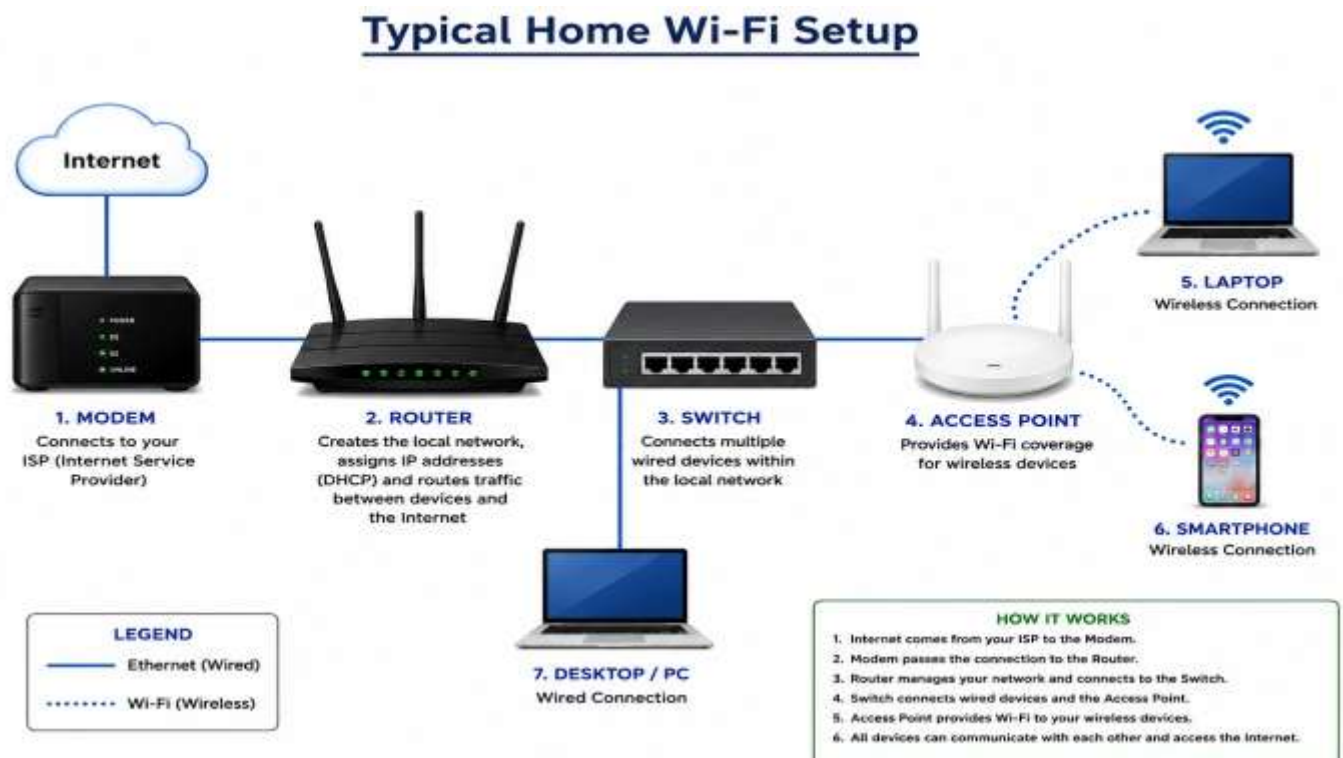


Comptia A+ Assignment

[A+ - Understanding and maintenance of Networks]

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

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1. Modem – Connects your home network to the Internet from the ISP.
2. Router – Manages the network and sends Internet traffic to devices.
3. Switch – Connects multiple wired devices together.
4. Access Point – Provides Wi-Fi connection to wireless devices.
5. Laptop – Connects to the network using Wi-Fi.
6. Smartphone – Connects wirelessly through Wi-Fi.
7. Ethernet Cable – Used for wired connections between network devices.
8. Wi-Fi – Provides wireless connectivity without cables.

2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

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Device	Main Function	Real-Life Example
Modem	Connects your home network to the Internet through your Internet Service Provider (ISP).	Used at home to receive Internet service from the ISP.
Router	Directs network traffic and shares the Internet connection between multiple devices.	Used at home to share Internet with laptops, phones, and TVs.
Switch	Connects multiple wired devices on the same local network.	Used in an office to connect computers and printers using Ethernet cables.
Access Point (AP)	Provides Wi-Fi access to devices and extends wireless network coverage.	Used in a school or office to provide Wi-Fi in different rooms.
Network Cable	Carries data between network devices using a wired connection.	Ethernet cable used to connect a computer to a router or switch.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

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1. Ethernet Cable (Cat5e/Cat6)



Common use: Used to connect computers, routers, switches, and other network devices in homes and offices

Source: [Wikimedia Commons – Cat5e Ethernet Cable](#)

2. Coaxial Cable



Common use: Commonly used for cable TV and broadband Internet connections.

3. Fiber Optic Cable



Common use: Used for **high-speed Internet and long-distance data transmission**, such as telecom networks and connections between buildings.

Source: [Wikimedia Commons – Cat5e Ethernet Cable](#)

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

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Feature	Router	Switch
Main Use	Connects different networks and provides Internet access.	Connects multiple devices within the same network.
Typical Location	Commonly used at home, offices, and businesses.	Commonly used in offices, schools, and data centres.
Data Handling	Routes data between different networks using IP addresses.	Forwards data between devices using MAC addresses.
Internet Connection	Usually connects the local network to the Internet.	Normally does not directly provide Internet access.
Devices Connected	Connects networks and devices such as computers, phones, and smart TVs.	Mainly connects wired devices such as computers, printers, and servers.
Example	A home router shares one Internet connection with multiple devices.	An office switch connects many computers using Ethernet cables.